A Search for Technetium, Promethium and Short-Lived Actinides in Nearby Sun-like Stars

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Abstract

Technetium, promethium and the actinides from protactinium to curium have no isotope that outlives a star, so their lines in a sun-like star would mean that atoms were added to it recently, for instance as fission waste disposed of by a civilization. We search 1413 archival HARPS, UVES, FEROS and ESPRESSO spectra of 274 F8 to K1 dwarfs within 50 pc for absorption at 50 lines of Tc I, Pm II, Pa II, Pu II, Am II and Cm II. Each star's stacked spectrum is fitted at each line by its six most similar stacks, and the element's combined absorption is compared with the same measurement at 400 random wavelengths of the same star. Of 2262 star-element tests, 9 fall below $p = 0.01$ where 22.6 are expected, and each of the nine takes its absorption from a stellar blend that other stars share or from a feature wider than an unresolved line. The median limits on the equivalent width of an element's strongest line run from 0.4 to 5.3 mÅ, and lines injected at those strengths are recovered at $p = 0.0010$. With curves of growth from Korg, the HARPS and ESPRESSO limits for solar-type stars are $A(\mathrm{Tc}) < 0.69$ and $A(\mathrm{Pm}) < -0.86$, near the solar abundance of thorium. Mixed through a solar convective envelope, they amount to 1.7×10^{19} kg of ^{99}Tc and 7×10^{17} kg of ^{147}Pm, the steady-state waste of 6×10^{21} W and 3.5×10^{25} W of fission power; the technetium alone is the product of fissioning 7×10^{20} kg of ^{235}U, several thousand times the uranium of the Earth, so the search constrains disposal on a scale far beyond a planet's fuel, as dilution through a sun-like star's convective envelope requires.

1 Introduction

Technetium, promethium and the actinides protactinium, plutonium, americium and curium have no isotope that lasts the life of a star. The longest-lived isotopes decay in 4.2 Myr (technetium), 17.7 yr (promethium), 33 kyr (protactinium), 81 Myr (plutonium), 7.4 kyr (americium) and 16 Myr (curium), so a star a few billion years old keeps none of the atoms of these elements it was born with, and their lines in its spectrum mean the atoms were made or delivered recently. Merrill's detection of technetium in S stars [1] showed nucleosynthesis inside evolved stars, but a main-sequence dwarf makes none of these elements.

That absence makes them a possible technosignature. Drake and Shklovskii suggested that a civilization could announce itself by salting its star with a short-lived isotope, and Sagan cautioned that normal and peculiar stellar spectra would have to be understood far better before an unusual atom could be read as a sign of intelligence [2]. Whitmire and Wright proposed that a civilization disposing of fission waste in its star would leave such lines without intending to, and argued that convective dilution and stellar lifetimes favour stars of types A5 to F2 [3, 4]. The only stars in which lines of these elements have been reported are chemically peculiar: promethium in Przybylski's star (HD 101065), HD 965 and HR 465 [5, 6], and short-lived actinides in Przybylski's star [7], where a synthesis of the Tc I 4297 Å line limits technetium to $\log \epsilon(\mathrm{Tc}) < 2.5$ [8] and a re-examination of three Pm II lines finds them too blended for promethium to be claimed [9]. Those reports remain disputed, since the spectra of such stars are crowded with lines of the rare earths, and natural

sources have been proposed, from nuclear reactions in the atmosphere [11] to the decay of long-lived superheavy nuclei [12].

The sun-like stars nearest the Sun, whose planets are the first targets of any search for life, have not been searched for these elements. This paper does so with the archival high-resolution spectra of the European Southern Observatory: we measure the absorption at the lines of technetium, promethium, protactinium, plutonium, americium and curium in the stacked spectra of nearby F8 to K1 dwarfs from four spectrographs, compare it with the same measurement at random wavelengths of the same star, and turn each star's sensitivity into an equivalent width, an abundance and a mass of material mixed through the star's convective envelope.

1.1 Contributions

1. A search of 274 nearby F8 to K1 dwarfs for six elements with no long-lived isotope in 1413 spectra from four spectrographs, with each star's limit on the equivalent width, the abundance and the mass of each element.
2. A measurement against comparison stacks with a per-star empirical null, whose limits follow from the measured line scores and are confirmed by injection.
3. Curves of growth of Tc I and Pm II in dwarf atmospheres of 5100 to 6500 K, and the conversion of the limits into masses of fission products and the fission power that would sustain them.

2 Data

The spectra come from the ESO Science Archive and four spectrographs: HARPS [17], UVES [18], FEROS [19] and ESPRESSO [20]. The UVES, FEROS and ESPRESSO samples are drawn from Gaia DR3 [14]: every source with parallax above 20 mas, $0.62 < G_{\mathrm{BP}} - G_{\mathrm{RP}} < 1.0$, $G < 10$, a renormalised unit weight error below 1.4 and declination below $+30°$, which selects the 1699 single F8 to K1 dwarfs within 50 pc that the southern telescopes reach [16]. For each spectrograph we count the one-dimensional spectra within $30''$ of each star, take the 150 (UVES), 92 (FEROS) and 94 (ESPRESSO) nearest stars with at least two of them and a Gaia radial velocity [15], and download up to four spectra of each covering 4220–4310 Å, those of highest signal-to-noise ratio first. The HARPS sample was drawn first and on narrower cuts: the 97 stars within 30 pc with $G < 9$, declination below $+20°$, the same colour and astrometric cuts, at least five HARPS spectra, a Gaia radial velocity and fewer than 200 references in SIMBAD (8 to 194), with up to six spectra each, highest signal-to-noise ratio first and one a night. A file found within $30''$ of two stars, the members of a wide pair, belongs to the one nearest the pointing in its header. Table 1 lists the samples; together they hold 1413 spectra of 274 distinct stars at 6.8 to 47.3 pc (median 26.2 pc).

3 Method

Stacks. Each spectrum is placed on a grid of 0.01 Å from 4100 to 5600 Å in air wavelengths and the barycentric frame, using the pipeline's frame where the file declares one and otherwise a barycentric correction for the observatory at mid-exposure. It is divided by its running upper envelope, the 92nd percentile in blocks of 3 Å, and a star's spectra are median-stacked. A stack whose scatter between adjacent samples exceeds 10 per cent at 4150–4550 Å is dropped. The stack is moved to the star's rest frame by the velocity that puts the cores of up to 30 strong Fe I, Ca I, Mg I and Ba II lines at their laboratory wavelengths. These velocities follow the Gaia radial velocities with a robust scatter of 0.11 km s^{-1} for HARPS and ESPRESSO, 0.25 km s^{-1} for FEROS and 0.53 km s^{-1}

Table 1: The samples. R is the resolving power assumed for the matched filter, and coverage the median span of a stack on the 4100–5600 Å grid. Eligible stars are those meeting the selection, of which the nearest are downloaded (all 97 for HARPS), with the number of spectra downloaded. Measured stars have a stack whose best comparison stack correlates above 0.9. S/N is the median signal-to-noise ratio of a stack per 0.01 Å sample at 4150–4550 Å.

Spectrograph	R	Coverage (Å)	Eligible	Stars	Spectra	Measured	S/N
HARPS	115,000	4100–5600	97	97	582	95	72
UVES	45,000	4100–4524	361	150	252	116	92
FEROS	48,000	4100–5600	138	92	283	80	96
ESPRESSO	140,000	4100–5600	102	94	296	86	63

Table 2: The lines measured and their weights relative to the element's strongest line, $\lambda^2 gf \exp(-E_{\rm low}/kT)$ at 5800 K. NIST lists all 23 Pu II lines of the range at the same arc intensity.

Ion	Air wavelengths (Å) and weights	gf
Tc I	4238.19 (0.45), 4262.27 (0.71), 4297.06 (1.00)	Palmeri et al. [21]
Pm II	4137.95 (0.59), 4157.86 (0.88), 4186.03 (1.00), 4297.78 (0.80), 4381.88 (0.12), 5546.08 (0.13), 5556.88 (0.11), 5561.73 (0.18), 5576.02 (0.20)	Fivet et al. [6]
Pa II	4217.23 (0.99), 4248.08 (1.00), 4176.18 (0.29), 4291.34 (0.31), 4601.43 (0.35)	NIST arc intensity
Pu II	4105.95 (0.82), 4129.93 (0.83), 4141.20 (0.83), 4159.39 (0.84), 4178.28 (0.85), 4189.90 (0.85), 4190.06 (0.85), 4196.20 (0.86), 4229.77 (0.87), 4254.76 (0.88), 4273.34 (0.89), 4289.08 (0.89), 4337.18 (0.91), 4352.71 (0.92), 4379.91 (0.93), 4385.35 (0.93), 4393.93 (0.94), 4441.65 (0.96), 4468.54 (0.97), 4472.79 (0.97), 4493.78 (0.98), 4504.91 (0.99), 4536.15 (1.00)	NIST arc intensity
Am II	4509.45 (0.97), 4575.59 (1.00), 4324.57 (0.36), 4441.36 (0.38), 4699.70 (0.42), 4872.22 (0.45), 4188.12 (0.17), 4593.31 (0.20), 5020.96 (0.24)	NIST arc intensity
Cm II	4207.66 (1.00)	NIST arc intensity

for UVES, about median offsets of +0.25 to +0.67 km s^{-1}. Nine stacks differ from Gaia by more than 3 km s^{-1}: four are spectroscopic binaries or stars whose Gaia velocities vary (HD 26491, τ^1 Eri, AF Lep and HD 189245), and five hold FEROS spectra from 2003 December 7 and 8, whose reductions are offset by 4 to 6 km s^{-1}. Because each stack's frame comes from its own lines, neither affects the measurement.

Lines. Table 2 lists the lines. Each line is weighted by its expected strength in a 5800 K atmosphere, $\lambda^2 gf \exp(-E_{\rm low}/kT)$ for the dominant ion, with the Tc I oscillator strengths of Palmeri et al. [21] and the Pm II values of Fivet et al. [6]. No oscillator strengths have been measured or computed for most lines of the singly ionized actinides, so their relative intensities in the arc spectra compiled by NIST [22] stand in for gf and every lower level is taken as the ground. Arc intensities depend on the source's excitation and on the detector as well as on gf, so the actinide weights are uncertain by factors of a few, which shifts strength between the lines of a combination but leaves each line's own measurement unchanged; the actinide results are therefore reported as equivalent widths of the strongest line and not as abundances. Lines within 3 Å of a Balmer line, and lines weaker than a tenth of the element's strongest, are left out.

Measurement. At each line we fit the ±8 Å around it by a non-negative combination of the six stacks of the same spectrograph that correlate best with the star's, together with a signed linear continuum and signed first and second derivatives of the fitted model, which absorb small differences of velocity and line width; the ±0.25 Å around the line is left out of the fit. A matched filter with the spectrograph's Gaussian line profile, applied to the residual at the line, gives the strength z of any extra absorption in units of the residual's robust scatter and its equivalent width. Stars whose best comparison stack correlates below 0.9 are not measured. The comparison stacks carry the star's own blends and the spectrograph's artefacts, so the residual at a line is what the star has that stars like it lack.

Score and null. An element's lines are combined into $z_c = \sum_i w_i z_i / (\sum_i w_i^2)^{1/2}$, where w_i are the weights. The same measurement at 400 random wavelengths of the same star, each at least 1 Å from any line of the six elements and from the Balmer lines and 10 Å inside the stack's coverage, as the lines measured are, is the null: p is the fraction of 20,000 combinations of null values, drawn with the same weights, that fall at or below z_c, and for a single line its rank among the null values. The null carries whatever the comparison stacks fail to model at random places of the spectrum, and one null per star serves all six elements. A line whose scores scatter across a spectrograph's stars by more than 1.5 times the median scatter of the stars' nulls lies on a stellar feature that the comparison stacks do not follow and is left out of that spectrograph's scores.

Limits. A line injected into a stack is narrower than the part of the window left out of the fit, since four standard deviations of the line profile span less than 0.25 Å for all four spectrographs, so it leaves the fitted model unchanged and moves each z_i in proportion to its equivalent width. A star's limit is therefore the equivalent width of the element's strongest line, the others in proportion to their weights, at which z_c would have reached the $p = 0.001$ point of the star's null, an absorber removing a fraction of the light at its line in proportion to the stack's flux there. For curium, whose one line gives a null of 400 values, the limit places the line at the most extreme of them. The limit follows from the measured line scores without refitting, and injection checks it: lines of each element injected at the limit into the stacks of twelve stars of each spectrograph, 288 injections in all, are recovered at a median p of 0.0010 for the elements with more than one line, with 80 per cent of them between 0.0008 and 0.0013.

Abundances. We convert a limit into an abundance $A(\mathrm{X}) = \log_{10}(N_\mathrm{X}/N_\mathrm{H}) + 12$ with curves of growth computed by Korg [23] in MARCS model atmospheres [24] of solar metallicity, from 5100 to 6500 K with $\log g$ from 4.55 to 4.35, with hydrogen lines left out of the synthesis so that Balmer wings do not enter the equivalent width. A star's temperature comes from its $G_\mathrm{BP} - G_\mathrm{RP}$ colour through the dwarf sequence of Pecaut and Mamajek [16]. A metallicity of ±0.3 dex changes the Tc I conversion by less than 0.01 dex and the Pm II one by 0.09 dex. To turn an abundance into a mass we mix the element through a solar convective envelope of $0.024\,M_\odot$ with a hydrogen mass fraction of 0.74 [26], and to turn the mass into a supply rate we hold it against decay, which for fission products of ^{235}U at 200 MeV a fission and cumulative yields of 6.13 per cent (^{99}Tc, half-life 2.111×10^5 yr) and 2.23 per cent (^{147}Pm, 2.62 yr) [27] gives a fission power.

4 Results

No element departs from chance. Table 3 gives each element's result in each spectrograph. Of the 2262 star-element tests, 186 fall below $p = 0.1$, 93 below 0.05, 9 below 0.01 and 1 below 0.001, where 226, 113, 22.6 and 2.3 are expected; nine or fewer below 0.01 has a Poisson probability of 0.001. The p values gather toward the middle of their range and thin at both ends (Figure 1), so the nulls, and the limits drawn from them, are conservative. The scatter criterion makes them so:

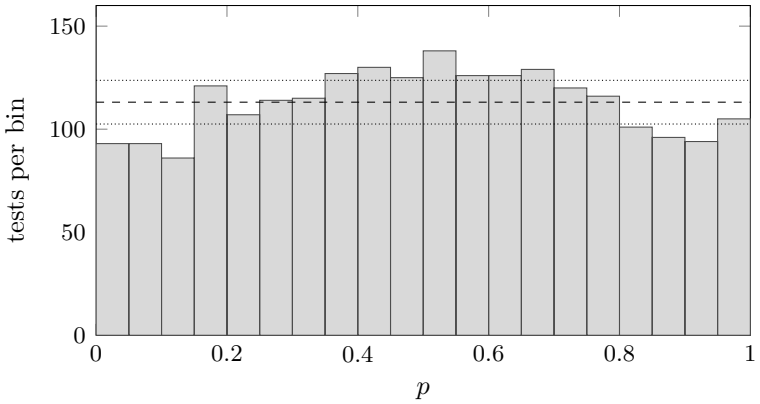


Figure 1: The p values of the 2262 star-element tests in bins of 0.05, against the 113 that a calibrated null puts in each bin (dashed) and its Poisson scatter (dotted).

Table 3: Results by element and spectrograph. N is the number of stars measured, $p_{\min}$ the lowest p among them, and W the median limit (mÅ) on the equivalent width of the element's strongest line, the others in proportion to their weights, at $p = 0.001$ (for curium, beyond every value of the star's null).

	HARPS			UVES			FEROS			ESPRESSO		
Ion	N	$p_{\min}$	W	N	$p_{\min}$	W	N	$p_{\min}$	W	N	$p_{\min}$	W
Tc I	95	0.038	1.59	116	0.029	3.80	80	0.061	5.32	86	0.063	1.60
Pm II	95	0.020	0.94	116	0.029	2.62	80	0.034	3.35	86	0.019	0.98
Pa II	95	0.016	1.43	116	0.003	2.87	80	0.017	4.30	86	0.002	1.30
Pu II	95	0.002	0.42	116	0.005	1.12	80	0.018	1.26	86	0.030	0.38
Am II	95	0.002	0.57	116	0.0001	3.40	80	0.013	1.84	86	0.016	0.53
Cm II	95	0.068	1.72	116	0.017	3.59	80	0.037	5.30	86	0.087	1.79

it leaves out the lines whose scores spread most across a spectrograph's stars, 13 to 19 per cent of whose scores lie beyond three times a star's robust null scatter, and the lines it keeps lie there in 1.3 to 2.8 per cent of cases where the random wavelengths of the nulls do in 1.5 to 3.6 per cent. The tests are not independent, since one null serves a star's six elements. The lowest, $p = 1.0 \times 10^{-4}$ for americium in the G3 V star HD 19467 (UVES), would be met by chance in 0.23 of such sets of tests. Each of the nine takes most of its absorption from one line. At eight of those lines at least six other stars of the same spectrograph absorb at least half as deeply, the mark of a stellar blend whose strength varies among stars: Am II 4324.57 Å, which carries HD 19467's score at $z = -19$, reaches below $z = -11$ in five other UVES stars. The ninth, plutonium in the K1 V double star HD 7895 (HARPS, $p = 0.0021$), absorbs across 4536.10–4536.45 Å, about 0.35 Å and several times the width of an unresolved line.

Lines that follow the comparison stacks poorly. Some plutonium lines lie on lines of ionized metals, whose strength follows a star's gravity, metallicity and alpha enhancement more closely than its comparison stacks do: Pu II 4468.54 Å on Ti II 4468.49 Å and 4385.35 Å on Fe II 4385.38 Å. The scatter criterion leaves out six Pu II lines in HARPS (4254.76, 4337.18, 4385.35, 4393.93, 4468.54 and 4472.79 Å), three in UVES and two each in FEROS and ESPRESSO, and no line of the other elements. Without it, the 0.97-weight line at 4468.54 Å alone gives Gaia DR3 5730371484020807808, a star moving at 119 km s^{-1}, a plutonium score of $z = -17$ against its comparison stacks.

Limits. The median limit on Tc I 4297.06 Å is 1.6 mÅ with HARPS and ESPRESSO and 3.8

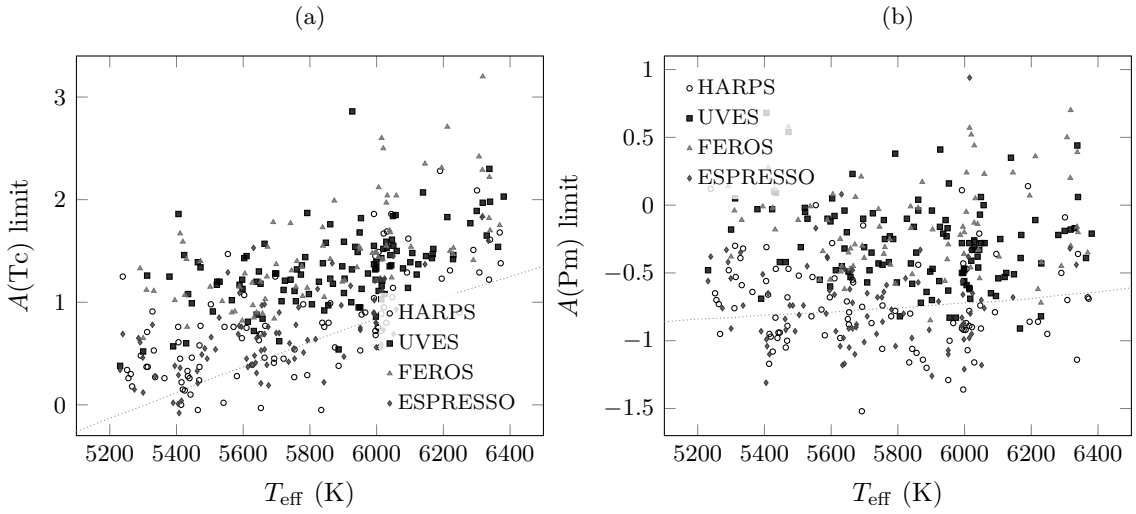


Figure 2: Each star's abundance limit against its temperature: (a) technetium from Tc I 4297.06 Å and (b) promethium from Pm II 4186.03 Å. The dotted curves mark the abundance at which the line's equivalent width is 1 mÅ. The curves of growth run from $A = -1$ to 3.5; one FEROS technetium limit lies above the top of that grid and is drawn at it.

and 5.3 mÅ with UVES and FEROS, whose resolving powers are 2.4 to 3 times lower and whose UVES stacks mostly hold one spectrum. For Pm II 4186.03 Å the median limits are 0.9 and 1.0 mÅ with HARPS and ESPRESSO, and for plutonium, whose limit combines up to 21 lines, 0.4 mÅ. Figure 2 shows each star's technetium and promethium limit as an abundance.

Abundances, masses and power. Tc I is a minority species whose ionization rises steeply with temperature, so 1 mÅ in the 4297 Å line corresponds to $A(\mathrm{Tc}) = -0.13$ at 5200 K, 0.61 at 5800 K and 1.25 at 6400 K, while the same equivalent width in Pm II 4186 Å, the dominant ion, corresponds to $A(\mathrm{Pm})$ between -0.84 and -0.64 across that range. The median limits are $A(\mathrm{Tc}) < 0.72$ and 0.66 with HARPS and ESPRESSO and 1.32 and 1.42 with UVES and FEROS, and the deepest is -0.08; for promethium they are -0.78, -0.76, -0.31 and -0.18, and the deepest -1.52. These are near the solar abundance of thorium, $A(\mathrm{Th}) = 0.02$ [25], and two dex below the technetium limit in Przybylski's star [8]. For the HARPS and ESPRESSO stars of 5600 to 6000 K, the median limits $A(\mathrm{Tc}) < 0.69$ and $A(\mathrm{Pm}) < -0.86$ correspond to 1.7×10^{19} kg of ^{99}Tc and 7×10^{17} kg of ^{147}Pm in a solar convective envelope. Holding them there against decay takes 5.6×10^{13} kg of technetium or 1.9×10^{17} kg of promethium a year, the products of fissioning ^{235}U at 5.6×10^{21} W or 3.5×10^{25} W, which is 0.09 of the Sun's luminosity. Put as fuel rather than power, 1.7×10^{19} kg of ^{99}Tc at a cumulative yield of 6.1 per cent is the product of 1.7×10^{45} fissions, 6.6×10^{20} kg of ^{235}U, where the Earth's mantle and crust hold about 8×10^{16} kg of uranium of all isotopes [10] and 6×10^{14} kg of ^{235}U: the limit lies three to six orders of magnitude above what fissioning a planet's uranium could put into its star, and the Pm II limit, set by a 2.6 yr half-life, correspondingly higher as a rate.

5 Related work

Technosignatures in stellar composition. The proposals of Drake and Shklovskii [2] and of Whitmire and Wright [3] are reviewed with other signatures of interstellar archaeology by Carrigan [4]; we find no earlier search of ordinary dwarfs for the elements they name.

Short-lived elements in stars. Technetium is the signature of the third dredge-up in asymptotic

giant branch stars, and its presence or absence separates S stars that make their own s-process elements from those that received them from a companion [1, 13]. Promethium and the short-lived actinides have been reported only in the chemically peculiar Ap stars Przybylski's star, HD 965 and HR 465 [5, 6, 7], whose strong magnetic fields and stratified atmospheres concentrate the rare earths in the layers that form their spectra; in Przybylski's star itself technetium is limited to $\log \epsilon(\mathrm{Tc}) < 2.5$ [8].

Surveys of nearby stars. The archival spectra used here were taken to measure radial velocities and abundances, and the abundance surveys built on them report the elements whose lines are strong in sun-like stars; none reports technetium, promethium or the actinides beyond thorium.

6 Conclusion

No line of technetium, promethium, protactinium, plutonium, americium or curium is found in the archival HARPS, UVES, FEROS and ESPRESSO spectra of 274 sun-like stars within 50 pc. Every low p has its absorption from a stellar blend that other stars share or from a feature wider than an unresolved line, and the limits reach abundances near the Sun's thorium abundance in the best spectra.

The limits apply to atoms in the photosphere at the epochs of the archival spectra, and the conversion into masses assumes the element mixed through a convective envelope as massive as the Sun's; a star's own envelope, thinner in F stars and thicker in K stars, scales the mass in proportion. In a sun-like star the envelope's mass turns small abundances into masses that no planetary industry reaches: the technetium limit is the waste of several thousand Earths' worth of uranium, so what the search excludes is disposal on the scale of a civilization that mines its whole planetary system, or that fissions its star's own uranium, sustained for longer than the products' half-lives. That is the dilution Whitmire and Wright foresaw, and the stars they preferred, A5 to F2 dwarfs whose thin envelopes dilute a thousand times less, are not in the sample, nor are the Ap stars in which diffusion concentrates heavy elements in the photosphere; both have archival spectra from the same spectrographs, and the same measurement applied to them is where a planetary-scale signal could be found.

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